

Percentage of quantities



1

Fraction	$\frac{1}{1}$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{5}$	$\frac{1}{8}$	$\frac{1}{10}$	$\frac{1}{20}$	$\frac{1}{25}$	$\frac{1}{50}$	$\frac{1}{100}$
Percentage	100%	50%	25%	20%	12.5%	10%	5%	4%	2%	1%

2

a $\frac{1}{4}$

b 2

3

a $\frac{3}{20} \times 160 = 24$

b $\frac{5}{2} \times 12 = 30$

c $\frac{4}{25} \times 275 = 44$

d $\frac{7}{4} \times 20 = 35$

4

a $0.9 \times 20 = 18$

b $2.2 \times 10 = 22$

c $0.6 \times 60 = 36$

d $0.21 \times 400 = 84$

5 Some possibilities, student responses may vary:

$$140 \times 0.25$$

$$140 \times \frac{25}{100}$$

$$(140 \times 5\%) \times 5$$

6

a 9

b 9

c 9

d 8

e 55

f 56

g 18

h 72

i 26

j 27

k 32

l 160

7

a 8 g

b \$12

c \$60

d 6 mL

e 55

f 14 km

g 30 L

h \$18

8

a
i 3

ii 7

iii 21

b
i 7

ii 7

iii 49

c
i 7

ii 7

iii 49

d
i 7

ii 8

iii 81



Percentage comparisons

9

- a 62% of 46 b 27% of 230 c 250% of 18 d 45% of 180
e 29% of 13 f 36% of 48

10

- a Larger b Smaller. c i 49.2 ii 98.4

Applications

11 152

12

- a 2910 b 2340 c 5250

13

- a 3500 people b 385 children

14 2625 mg